

VocaLens — print day

Two jobs, different materials. **Start the PETG first** — it is the long one, and the TPU can run after it or on a second machine.

JOB 1 · PETG · the two pods~3 h

File	FINAL_petg_plate_HEADERS.stl — your XIAO has 2×7 headers soldered on, so this is the one. This build assumes the amplifier is wired WITHOUT sockets — wires soldered straight to its pads. That is what makes the rear pod 10mm slimmer. If your amp still has jumper sockets pushed onto pins, this pod will not close over it: print FINAL_petg_plate_AMP SOCKETS.stl instead.
What is on it	4 parts: front pod base, front pod lid, rear pod base, rear pod lid
Material	PETG (PLA is acceptable if PETG is not free — it is stiffer and more brittle, but it fits)
Layer height	0.2 mm
Infill	25 %
Perimeters / walls	3
Supports	OFF
Brim	ON — the pods are tall and narrow and can be knocked over
Plate size	105 × 108 × 27 mm — fits any printer at the venue

Why supports off: the plate is already arranged so nothing overhangs. The pods sit cavity-up and the lids sit flat face down. If your slicer warns about overhangs, something imported wrong — do not switch supports on to silence it.

JOB 2 · TPU · the four rings~40 min

File	Not measured your glasses arm yet? Print FINAL_tpu_rings_LADDER.stl first — four graduated sizes, ~40 min, smallest on the left. Fit each in turn and keep the smallest that goes on. Once you know the size: FINAL_tpu_rings.stl for four of it.
What is on it	4 copies of one small ring — 2 per pod, plus spares. All ring sizes carry an identical dovetail rail, so whichever size fits your frame still mates with the pods you already printed.
Material	TPU / flexible filament. This one cannot be substituted
Layer height	0.2 mm
Infill	20 %
Perimeters / walls	2 — not 3. The wall is only 1.2 mm; more perimeters make it too stiff to stretch
Print speed	20–25 mm/s. TPU jams if pushed fast
Retraction	Near zero (0.5 mm or less)
Supports	OFF
Plate size	68 × 10 × 12 mm (ladder 68 × 14 × 13 mm)

If the venue has no TPU, print the rings anyway in the softest filament available and expect them to grip poorly, then say so — but ask around first. The rings are the only part that has to flex, and a rigid ring will crack rather than stretch onto the arm.

WHEN THEY COME OFF THE BED — check in this order

1. **Slide a ring onto your glasses temple arm.** It should need a firm push and then stay put on its own.

If it is loose, go down one step on the ladder — or reduce `RING_IN_VERT` / `RING_IN_OUT` in `vocalens_ring.scad` by 0.5 mm and reprint one ring (10 minutes). **If it will not go on at all, go UP a step.** Smaller is the safer direction but it has a floor: TPU grips best at 20-30 % stretch, and past that it either refuses to fit or takes a permanent set and grips worse. Aim for roughly **75-85 % of the arm's measured section**.

2. **Slide a pod base onto a ring's dovetail rail**, along the arm. It should need real effort and should not rattle or lift off sideways.

If it will not go on at all, lower `DT_SQUISH` from 0.45 to 0.35. **If it rattles**, raise it to 0.55.

3. **Drop a lid onto its base.** The rib should sit inside the cavity and the lid should sit flush.
4. **Check the front lid has two openings:** a round hole near one end for the camera, and an 18 × 20 mm oval recess near the other for the touch pad, with a small hole through the middle of the oval.

The small hole is for the touch wire. Solder the wire to the **back** of the pad *before* gluing the pad into the recess — afterwards you cannot reach it, and nothing conductive should be exposed where a finger touches.

5. **Check the rear lid** has a ring of small holes — that is the speaker grille.
6. **Drop the amplifier into the rear pod before you glue anything.**

This is the one clearance the slim design spends. If the amp still has sockets on it, it will not go in — and you want to find that out now, holding both parts, not at midnight with wires already run.

TAKE WITH YOU

- ☐ Laptop with the STL files on it (or the GitHub repo open)
- ☐ USB stick or SD card — most venue printers will not take a network job
- ☐ **The glasses you are mounting to** — the ring fit test is the whole point and cannot be done later
- ☐ Your assembled XIAO + camera, to check it drops into the front pod cavity
- ☐ The speaker and amplifier, to check they drop into the rear pod cavity
- ☐ Superglue or double-sided tape for the touch pad

Print the TPU rings even if you are short on time. The pods are useless without them, and they are the fastest part on the list.